Review

2.6 Syllabus covered

- ☑ Binary numbers
- ☑ Generate minterms, maxterms, SOP canonical form and POS canonical forms and convert between them
- ☑ Understand and use the laws and theorems of Boolean Algebra
- ☑ Perform algebraic simplification using Boolean algebra
- ☑ Simplification using K-maps
- ☑ Derive sum of product and product of sums expressions for a combinational circuit
- ☑ Convert combinational logic to NAND-NAND and NOR-NOR forms
- Hexadecimal, Sign-magnitude, One's-complement and Two's complement. Conversions between them.
- Design combinational circuits for positive and negative logic
- Design Hazard-free two level circuits and understand Hazards in multi-level circuits
- Compute noise margin of one device
- Describe how tri-state and open-collector outputs are different from totem-pole outputs.
- Different between and limitations of master-slave and edge-triggered flip-flops.
- Compute fan out and noise margin of one device driving the same time
- Know the differences and similarities between PAL, PLA, and ROMs and can use each for logic design
- Design combinational circuits using multiplexers and decoders
- Analyze a sequential circuit and derive a state-table and a state-graph
- Understand the difference between synchronous and asynchronous inputs
- Derive a state graph or state table from a word description of the problem
- Reduce the number of states in a state table using row reduction and implication tables
- Perform a state assignment using the guideline method

- Implement a design using JK, SR, D or T flip-flops
- Analyse and design both Mealy and Moore sequential circuits with multiple inputs and multiple outputs
- Convert between Mealy and Moore designs
- Partition a system into multiple state machines

2.6.1 Labs (not questioned in exams)

- Use computer tools to enter designs graphically and HDL
- Simulate designs using computer tools
- Use computer tools to program gate arrays logic and debug and test

Chapter 3

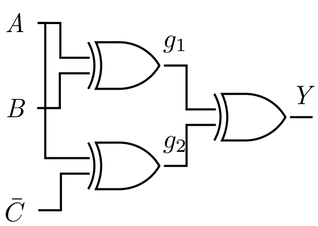
Sample midterm exam

Student Name: Student Email:

3.1 Instructions

- Time allowed is ~~50~~ 75 min minutes. (This sample exam might be lengthier than the actual exam. Same instruction will apply to the midterm.)
- In order to minimize distraction to your fellow students, you may not leave during the last 10 minutes of the examination.
- The examination is closed-book. One 8x11in cheatsheet is allowed.
- Non-programmable calculators are permitted.
- The maximum number of marks is 80, as indicated; the midterm examination amounts 10% toward the final grade.
- Please colored pen/pencils for K-maps, use a pen or heavy pencil to ensure legibility.
- Please show your work; where appropriate, marks will be awarded for proper and well-reasoned explanations.

Problem 3.1. *Consider the circuit below*



$g_1 = A \oplus B$

$g_1 = m_1 + m_2$

$= \bar{A}B + A\bar{B}$

$g_2 = \bar{A}\bar{C} + A\bar{\bar{C}} = \bar{A}\bar{C} + AC$

A	B	g_1	
0	0	0	
0	1	1	← m_1
1	0	1	← m_2
1	1	0	

By algebraic manipulation, prove or disprove that $Y = \bar{B}\bar{C} + BC$ *(10 marks).*

Problem 3.2. *Use the following 4-variable K-map for F (A, B, C, D), and find a minimal sum of products (SOP) expression for F (15 marks)*

$\overline{(\bar{A}\bar{C} + AC)}$ XNOR

$= (A+C)(\bar{A}+\bar{C})$

$= A\bar{A} + C\bar{C} + A\bar{C} + \bar{A}C$

$= 0 + 0 + A\bar{C} + \bar{A}C$

$= A\bar{C} + \bar{A}C$ ← XOR

$Y = g_1\bar{g}_2 + \bar{g}_1 g_2$

$= \underbrace{(\bar{A}B + A\bar{B})}_{XOR}\,\overline{(\bar{A}\bar{C} + AC)}$ XNOR

$+ \overline{(\bar{A}B + A\bar{B})}(\bar{A}\bar{C} + AC)$

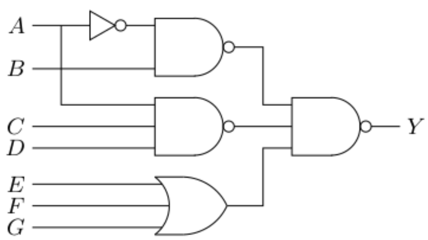
$= (\bar{A}B + A\bar{B})(A+C)(\bar{A}+\bar{C})$

$$\overline{(\bar A B + A\bar B)} = (A+\bar B)(\bar A + B)$$
$$= A\bar A + \bar B B + AB + \bar A\bar B$$
$$= AB + \bar A \bar B$$

$$Y = (\bar A B + A\bar B)(A\bar C + \bar A C)$$
$$+ (AB + \bar A\bar B)(\bar A\bar C + AC)$$
$$= A\overset{0}{\cancel{\bar B B}}\bar C + \bar A BC + A\bar B\bar C$$
$$+ ABC + \bar A\bar B\bar C + \overset{0}{A\cancel{\bar A}\bar B C}$$
$$= (A+\bar A)BC + (A+\bar A)\bar B\bar C$$
$$= BC + \bar B\bar C$$

CD \ AB	00	01	11	10
00	1	0	0	1
01	1	1	0	1
11	0	1	0	0
10	0	1	1	0

Problem 3.3. *Use boolean algebra to find an SOP expression for Y in the circuit below. Draw an equivalent circuit beside the given circuit (5 marks).*



Problem 3.4. *Consider the function Y given below.*

$$Y(A,B,C,D) = \sum m(0,3,5,7,8,14) + d(2,12,15)$$

1. *Draw a K-maps to derive a minimum SOP and POS expressions for Y . Indicate all essential prime implicants for Y or $\bar{Y}$ in your K-maps (20 marks).*
2. *Sketch a two-level NOR-NOR circuit for Y. Assume that A, B, C, and D are available in true end complimentary forms (5 marks).*
3. *Write Y in Product of sums (POS) canonical form (5 marks).*

Sum of Products

Problem 3.5. *Design a minimal SOP circuit to add two two-bit unsigned numbers. Denote the two bits of first number as A_1A_0 and the two bits of second number as B_1B_0. The result will be a 2-bit sum S_1S_0 and a carry C. Start with filling out the following truth table (3 example rows are provided) and then use K-maps to find minimal SOP for S_1, S_0 and a single carry bit C_1 (20 marks).*

$A_1 A_0$	00	01	10	11
$B_1 B_0$	00	01	10	11

$$\begin{array}{r} A_1 A_0 \\ + \;\; B_1 B_0 \\ \hline C \;\; S_1 S_0 \end{array}$$

Problem 3.4. *Consider the function Y given below.*

MSB → LSB

$$Y(A, B, C, D) = \sum m(0, 3, 5, 7, 8, 14) + d(2, 12, 15)$$

1. *Draw a K-maps to derive a minimum SOP and POS expressions for Y. Indicate all essential prime implicants for Y or $\bar{Y}$ in your K-maps (20 marks).*
2. *Sketch a two-level NOR-NOR circuit for Y. Assume that A, B, C, and D are available in true end complimentary forms (5 marks).*
3. *Write Y in Product of sums (POS) canonical form (5 marks).*

Sum of Products

Y (A across the two right-hand columns, B across the two middle columns, D beside the two middle rows, C beside the bottom two rows)

1 (0)	0 (4)	d (12)	1 (8)
0 (1)	1 (5)	0 (13)	0 (9)
1 (3)	1 (7)	d (15)	0 (11)
d (2)	0 (6)	1 (14)	0 (10)

EPI $= \bar{A}BD$

$Y = \bar{A}BD + \bar{A}\bar{B}C + \bar{B}\bar{C}\bar{D} + AB\bar{D}$

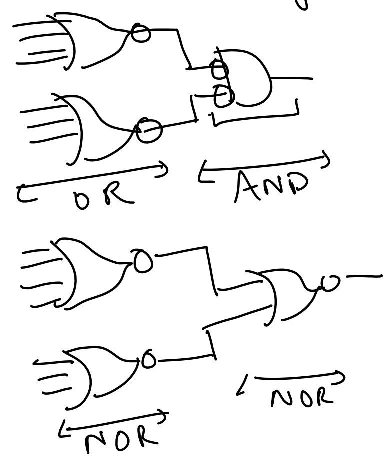
$\bar{Y}$

0	1	d	0
1	0	1	1
0	0	d	1
d	1	0	1

① The process of grouping:
- Finding all Prime implicants

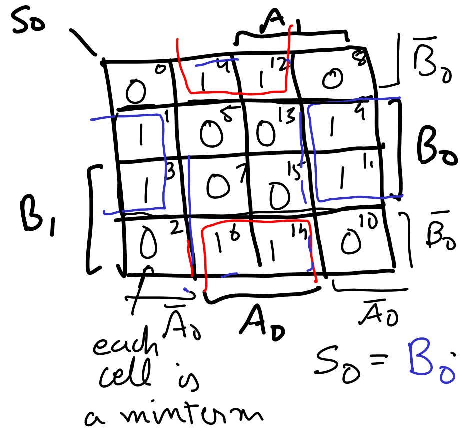
② Finding the minimal cover (finding fewest PI that cover all 1's)

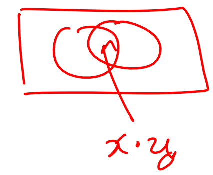
Product of sums form



MSB

A_1	A_0	B_1	B_0 (LSB)	C_1	S_1	S_0
0	0	0	0	0	0	0
0	0	0	1	0	0	1
0	0	1	0	0	1	0
0	0	1	1	0	1	1
0	1	0	0	0	0	1
0	1	0	1	0	1	0
0	1	1	0	0	1	1
0	1	1	1	1	0	0
1	0	0	0	0	1	0
1	0	0	1	0	1	1
1	0	1	0	1	0	0
1	0	1	1	1	0	1
1	1	0	0	0	1	1
1	1	0	1	1	0	0
1	1	1	0	1	0	1
1	1	1	1	1	1	0



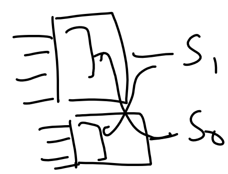


$$S_0 = B_0 \cdot \overline{A_0} + A_0 \cdot \overline{B_0}$$

Restricted

- 4-5 inputs
- two level circuits
  - → Sum of products
  - → Product of sums
- Single output

Quine McCluskey



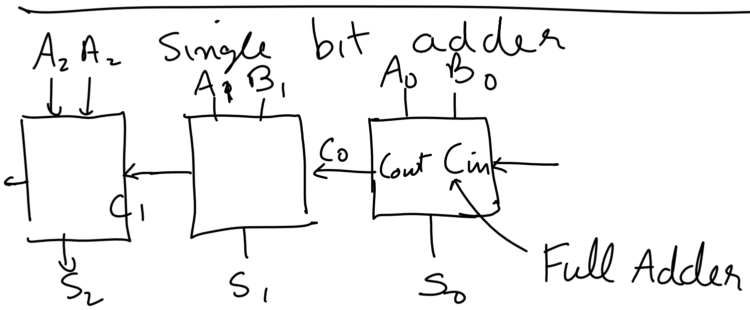
$\overline{1\ 0\ 0} \leftarrow 4$

$A_1 A_0$		0	0
$B_1 B_0$		0	0
	0	0	0
	C_1	S_1	S_0

1
0 1
0 1
0 1 0

0 1
1 0
1 1

1 1
0 1 ← 1
1 1 ← 3



A_0	B_0	Cin	Cout	S_0
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	1	0
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

S_0		A_0		
	0	1	0	0
C_{in}	1	0	1	0
		B_0		

C_{out}			A_0	
	0	0	1	1
C_{in}	0	1	1	1
		B_0		

$$S_0 = \bar{A}_0 \bar{B}_0 C_{in} + \bar{A}_0 B_0 \bar{C}_{in} + A_0 B_0 C_{in}$$

$$C_{out} = A_0 + B_0 C_{in}$$

Chapter 4

Muxes and Decoders

Multiplexers

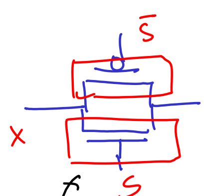
Building blocks of combinational circuits that you see repeatedly.

4.1 Objectives

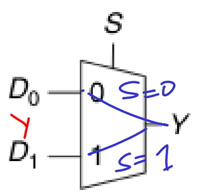
1. Design combinational circuits using multiplexers and decoders

4.2 Design combinational circuit using multiplexers [1, Section 2.8.1]

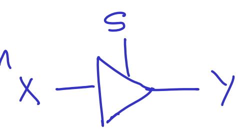
4.2.1 Review: 2to1 Multiplexer (MUX)



CMOS complementary MOSFET



Tristate Buffer

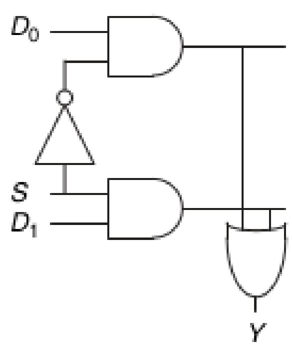


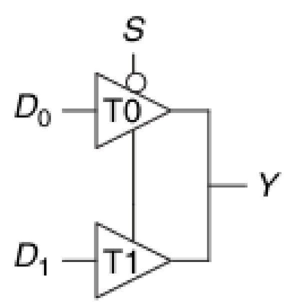
if S==1
then Y = X
else
Y = floating

S	X	Y
0	0	z
0	1	z
1	0	0
1	1	1

} floating

S	D_1	D_0	Y
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1



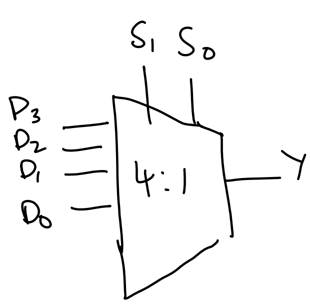


$Y = D_0\bar{S} + D_1 S$

Tristate Buffer

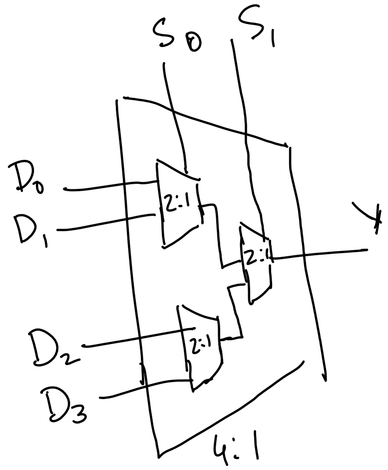
4.2.2 Wider multiplexers

Draw the symbol for a 4:1 MUX, an 8:1 MUX and a 2^N : 1 MUX and write corresponding Boolean expressions.



S_1	S_0	Y
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

$$Y = \underbrace{(\bar{S}_1\bar{S}_0)}_{m_0}D_0 + \underbrace{(\bar{S}_1 S_0)}_{m_1}D_1 + \underbrace{(S_1\bar{S}_0)}_{m_2}D_2 + \underbrace{(S_1 S_0)}_{m_3}D_3$$

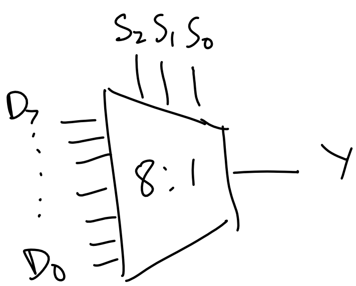


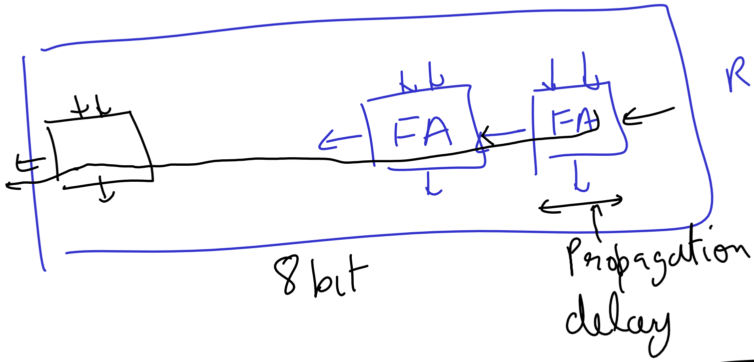
if ($S_1 S_0$ = 00)

.
.
.
.

```
if (S1 == 0){
    if (S0 == 0){
        Y = D0
    }else{
        Y = D1
}else{ if (S0 == 0){
        Y = D2
    }else{ Y = D3
    }
```

8:1 MUX





Ripple Adder

↳ High propagation delay

Lookahead carry adder

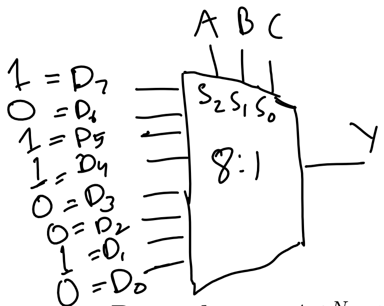
$\underbrace{\hspace{10em}}$

8 x Propagation delay

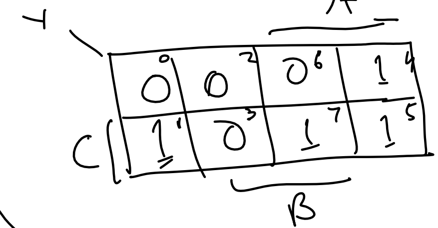
circuit timing { 8 x Propagation delay, Contamination delay }

3 inputs   3 selection bits

Example 4.1. *Design a circuit for* $\cancel{Y = A\bar{B} + \bar{B}\bar{C} + \bar{A}BC}$ *using a 8:1 MUX.*



A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

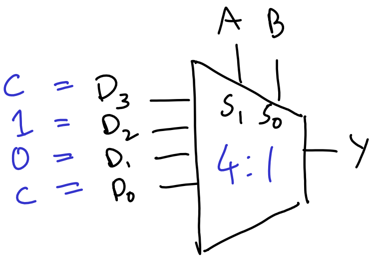


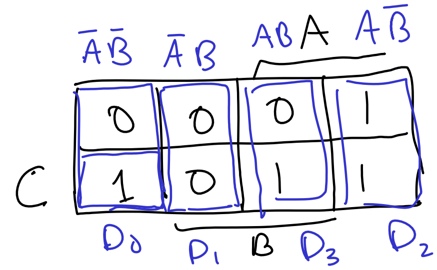
$Y = A\bar{B} + \bar{B}C + ABC$

Remark 4.1. *A* $2^N : 1$ *MUX can be used to program any N-input logic function.*

Example 4.2. *Design a circuit for* $Y = A\bar{B} + \bar{B}\bar{C} + \bar{A}BC$ *using a 4:1 MUX and NOT gates only.*

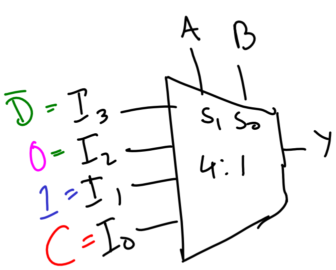
3 inputs   2 selection bits



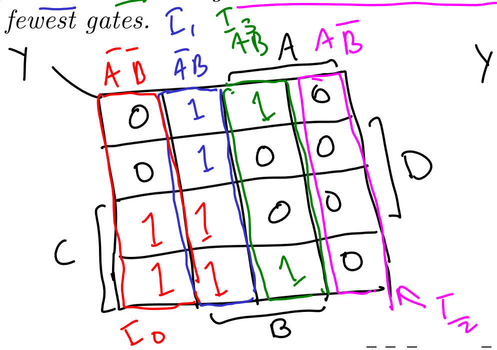


Remark 4.2. *A* $2^{N-1} : 1$ *MUX can be used to program any N-input logic function, if we use literals on the input side.*

Example 4.3. *Design a circuit for* $Y = \bar{A}C + \bar{A}B + B\bar{D}$ *using a 8:1 MUX and NOT gates only. Also design using 4:1 MUX and other gates.* fewest gates.



Repeat this process with AC connect to selection bits, BD connected to Input lines



$$Y = (\bar{A}\bar{B})I_0 + (\bar{A}B)I_1 + (A\bar{B})I_2 + (AB)I_3$$

Problem 4.1. *(15 marks [1, Ex-2.42]) Implement the function* $Y = BC + \bar{A}\bar{B}\bar{C} + B\bar{C}$ *using*

1. *an 8:1 multiplexer*
2. *a 4:1 multiplexer and no other gates*
3. *a 2:1 multiplexer, one OR gate, and an inverter*

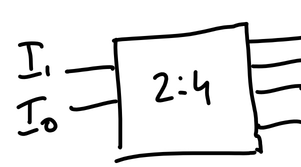
4.3 Encoders and Decoders

Example 4.4. *Draw the symbol and the truth table for 2:4 decoder. Also write the logic expressions.*

Decoders decode binary number to its one-hot encoding

I_1	I_0	O_3	O_2	O_1	O_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0

Binary number → one-hot encoding



$O_3 = m_3 = I_1 I_0$

$O_2 = m_2 = I_1 \bar{I_0}$

$O_1 = m_1 = \bar{I_1} I_0$

$O_0 = m_0 = \bar{I_1}\bar{I_0}$

Minterms are product expressions that contain all literals and result in a 1 for only 1 row of the truth table

A	B	C	f
			0
			0
			0
1 ·	1 ·	0 →	1
			0
			0
			0

$f = m_6$

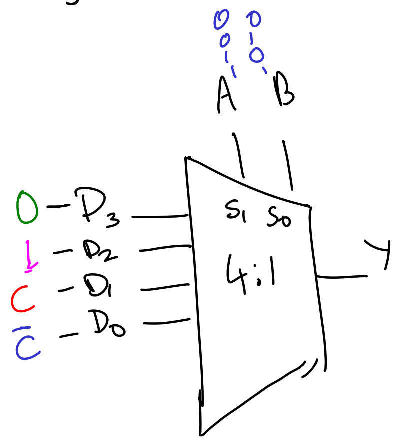
$m_6 = A\ B\ \overline{C}$

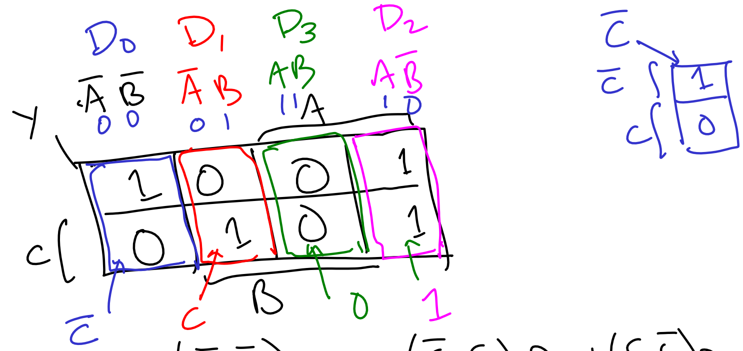
A	B	C	f
			0
			0
			0
1 ·	0 ·	1	1
			0
			0
			0

$f = m_5 = A \cdot \overline{B} \cdot C$

$$Y = A\bar{B} + \bar{B}\bar{C} + \bar{A}BC$$

Design a circuit for this expression using a 4:1 MUX

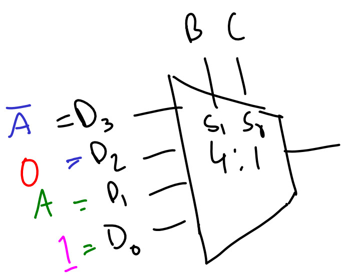


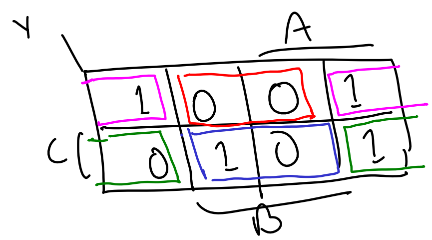


$$Y = (\bar{S}_1\bar{S}_0)D_0 + (\bar{S}_1 S_0)D_1 + (S_1\bar{S}_0)D_2 + (S_1 S_0)D_3$$

Repeat this design process with BC as the selection bits, with A connected to data lines.

$$Y = A\bar{B} + \bar{B}\bar{C} + \bar{A}BC$$



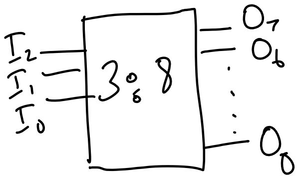


$$Y = (BC)D_3 + (B\bar{C})D_2 + (\bar{B}C)D_1 + (\bar{B}\bar{C})D_0$$

$S_1 S_0 = 00 = \bar{B}\bar{C} = m_0$

$BC = m_3$

Example 4.5. *Draw the symbol and the truth table for 3:8 decoder, 4:16 decoder and $N : 2^N$ decoder. Also write the logic expressions.*

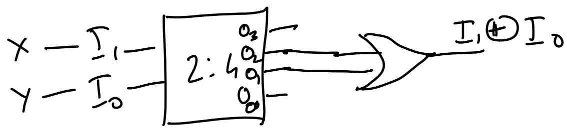


$O_7 = m_7 = I_2 I_1 I_0$

$O_6 = m_6 = I_2 I_1 \bar{I}_0$

Example 4.6. *Design a circuit for a XOR gate using a 2:4 decoder and an OR gate.*

$X \oplus Y = \underline{m_1} + \underline{m_2} = \bar{X} Y + X \bar{Y}$



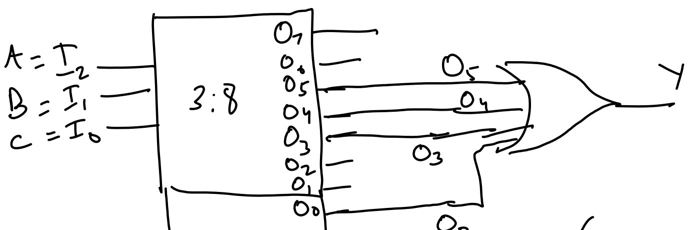
X	Y	$X \oplus Y$
0	0	0
0	1	1
1	0	1
1	1	0

$m_0 = O_0$
$m_1 = O_1$
$m_2 = O_2$
$m_3 = O_3$

Example 4.7. *Design a circuit for $Y = A\bar{B} + \bar{B}\bar{C} + \bar{A}BC$ using a 3:8 decoder and an OR gate.*

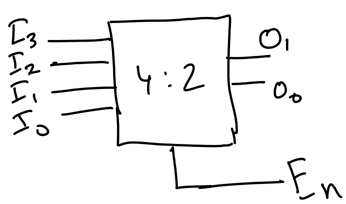
$Y = \sum m(0, 3, 4, 5)$

Y			A	A
	1^{0}	0^{2}	0^{6}	$1{+}1^{4}$
C	0^{1}	1^{3}	0^{7}	1^{5}
		B	B	



4.3.1 (Priority) Encoders (one-hot encoding → Binary number)

Example 4.8. *Draw symbol and truth table for a 4:2 priority encoder.*



$I_3 >$	$I_2 >$	$I_1 >$	I_0	O_1	O_0	E_n
0	0	0	0	d	d	0
0	0	0	1	0	0	1
0	0	1	x	0	1	1
0	1	x	x	1	0	1
1	x	x	x	1	1	1

≡

0	0	1	0	0	1	1
0	0	1	1	0	1	1

Example 4.9. *Draw symbol and truth table for a 8:3 priority encoder.*